Deep Learning for Two-Sided Matching (Report)

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1 Introduction

Matching problems arise in many everyday and institutional settings, including student–school assignment, worker–firm matching, and other two-sided allocation markets. Although pairing two sides may appear straightforward at first glance, the problem becomes nontrivial once resource limitations and heterogeneous preferences are taken into account: schools have limited seats, students value schools differently, and not all assignments are feasible or desirable. The theory of two-sided matching provides a rigorous framework to study such environments and has led to mechanisms that are widely used in practice (see Roth and Sotomayor [7] for a comprehensive survey in this literature).

In its most common form, a two-sided matching model requires three main modeling choices. First, one specifies the *market structure*: one-to-one matching (each agent matches with at most one partner) versus many-to-one matching (e.g., schools with capacities). Many-to-many environments exist as well [2], but are outside the scope of this project. Second, one selects the *desiderata* that the matching should satisfy. Beyond feasibility, typical criteria include stability, incentive properties such as strategy-proofness, welfare/efficiency objectives, fairness considerations, and distributional or capacity constraints (see, e.g., Aziz et al. [8] for a comprehensive survey on various distributional constraints). Third, one studies which desiderata are jointly attainable and the trade-offs induced by incompatibilities; for instance, certain combinations of stability and strategy-proofness are impossible in standard two-sided settings [3].

This project focuses on the algorithmic side of these trade-offs. Classical mechanisms are typically designed to guarantee a specific property (e.g., Deferred Acceptance for stability, Random Serial Dictatorship for strategy-proofness), yet they do not directly address the problem of optimizing multiple objectives when they cannot all be simultaneously satisfied. In such cases, the goal is naturally formulated as an optimization problem: choose a mechanism that maximizes a joint objective balancing several criteria. However, directly optimizing these criteria over the space of mechanisms is generally computationally intractable. Existing approaches include restricting preference domains to structured classes to recover compatibility [5], using optimization-based formulations such as linear programming [4], and more recently, leveraging machine learning to search for mechanisms exhibiting improved empirical trade-offs [6].

In the learning-based approach, the key steps are (i) defining differentiable surrogate objectives that quantify desiderata, (ii) choosing an architecture that can map discrete preference profiles to valid matchings while respecting inherent symmetries, and (iii) training a parameterized mechanism g^θ to optimize the desired trade-off. In this work, we explore deep-learning mechanisms for two-sided matching. We first study the one-to-one

setting and then consider extensions to many-to-one matching. Our primary objectives are *efficiency* and *fairness*, where fairness is defined as equality of outcomes between two groups and may conflict with efficiency notions such as the probability of receiving a first-choice assignment [9].

2 Problem Statement

Let $W = \{1, \ldots, n\}$ be the set of workers and $F = \{1, \ldots, m\}$ the set of firms. A one-to-one matching is any collection μ of pairs $(w, f) \in W \times F$ such that each worker and each firm appears in at most one pair. Let $\mathcal{B}$ denote the set of all such matchings. For any $(w, f) \in \mu$, we say that μ assigns worker w to firm f, and we use the notation $\mu(w) = f$ and $\mu(f) = w$. Agents who are not assigned a partner are treated as being matched with a special outside option $\perp$; in that case we write $\mu(w) = \perp$ (equivalently $(w, \perp) \in \mu$) or $\mu(f) = \perp$ (equivalently $(\perp, f) \in \mu$).

Each worker $w \in W$ has a strict ranking $\succ_w$ over $F \cup \{\perp\}$, and each firm $f \in F$ has a strict ranking $\succ_f$ over $W \cup \{\perp\}$. An agent deems any partner placed below $\perp$ unacceptable, meaning that the agent strictly prefers staying unmatched to being paired with such a partner. We write $f \succ_w f'$ to indicate that worker w prefers firm f to firm f', and analogously $w \succ_f w'$ for a firm's preferences. Let $\mathcal{P}$ be the set of all preference profiles; a profile is denoted by

$$\succ = (\succ_1, \ldots, \succ_n, \succ_{n+1}, \ldots, \succ_{n+m}) \in \mathcal{P},$$

where the first n components correspond to workers' orders and the remaining m components to firms' orders.

Given a matching μ, we evaluate its quality through two criteria: *efficiency* and *fairness*. Efficiency is understood in an ordinal sense, namely how well μ matches agents to highly ranked partners (e.g., the fraction of workers assigned to one of their top-k firms, and analogously for firms). Fairness is understood as equality of outcomes between two predefined groups of workers: a matching is considered fair when the average outcome quality is comparable across groups, and unfairness is quantified by the gap between these group averages. Finally, we require μ to be *individually rational (IR)*: no agent is matched to an unacceptable counterpart, i.e., every matched agent weakly prefers its assigned partner to remaining unmatched.

2.1 Theory of matching mechanisms

As discrete matchings are not appropriate for deep learning architectures, we work with continuous matchings or, more precisely, *randomized matchings*. Let

$$g : \mathcal{P} \to \Delta(\mathcal{B})$$

be a randomized matching mechanism mapping a preference profile to a distribution over matchings. For $w \in W$ and $f \in F$, we denote by $g_{wf}(\succ) \geq 0$ the *marginal* probability that worker w is matched to firm f under g given $\succ$. When allowing unmatched agents, we extend the indices to include the outside option $\perp$, i.e., we interpret $g_{w\perp}(\succ)$ (resp. $g_{\perp f}(\succ)$) as the probability that w (resp. f) remains unmatched.

In the one-to-one setting, we require the marginal feasibility constraints

$$\sum_{f' \in F \cup \{\perp\}} g_{wf'}(\succ) = 1 \quad \forall w \in W, \qquad \sum_{w' \in W \cup \{\perp\}} g_{w'f}(\succ) = 1 \quad \forall f \in F,$$

together with $g_{\perp\perp}(\succ) = 0$. Equivalently, $g(\succ)$ can be seen as an $(n+1) \times (m+1)$ nonnegative matrix whose row and column sums satisfy the above constraints.[1]

To convert a randomized matching to a discrete matching, we use the **Birkhoff–von Neumann theorem**, which states that any doubly stochastic matrix can be written as a convex combination of permutation matrices. In our context (ignoring $\perp$ for simplicity), $g(\succ)$ admits a decomposition

$$g(\succ) = \sum_{t=1}^{T} \alpha_t P^{(t)}, \qquad \alpha_t \geq 0, \;\; \sum_{t=1}^{T} \alpha_t = 1,$$

where each $P^{(t)}$ is a permutation matrix corresponding to a deterministic matching. We can therefore sample a discrete matching by drawing $P^{(t)}$ with probability α_t. For the case where the matrix is not square, we artifically complete it with zeros to make it square.

To encode ordinal preferences, each agent's ranking is mapped into a vector with a constant offset between successive positions. Let $p^{\succ} = (p^{\succ}_w)_{w \in W}$ and $q^{\succ} = (q^{\succ}_f)_{f \in F}$ encode workers' and firms' orders, with

$$p^{\succ}_{w\perp} = 0, \qquad q^{\succ}_{\perp f} = 0.$$

Examples:

$$w_1 : f_1 \succ f_2 \succ \perp \succ f_3 \quad \Rightarrow \quad p^{\succ}_{w_1} = (2/3,\ 1/3,\ -1/3),$$
$$w_1 : f_1 \succ f_2 \succ \perp \succ f_3 \succ f_4 \quad \Rightarrow \quad p^{\succ}_{w_1} = (2/4,\ 1/4,\ -1/4,\ -2/4).$$

Formally, for worker $i \in W$ and firm $j \in F$,

$$q^{\succ}_{if} = \frac{1}{n}\,\mathbf{1}\{i \succ_f \perp\} + \sum_{j'=1}^{n} \Big(\mathbf{1}\{i \succ_f j'\} - \mathbf{1}\{\perp \succ_f j'\}\Big), \tag{3}$$

$$p^{\succ}_{wj} = \frac{1}{m}\,\mathbf{1}\{j \succ_w \perp\} + \sum_{j'=1}^{m} \Big(\mathbf{1}\{j \succ_w j'\} - \mathbf{1}\{\perp \succ_w j'\}\Big). \tag{4}$$

We now adapt the notions of fairness and efficiency to randomized matchings.

2.1.1 Fairness.

Let W denote the set of workers and F the set of firms. We define a binary group label function

$$\ell : W \to \{A, B\}, \qquad W_A \coloneqq \{w \in W : \ell(w) = A\}, \qquad W_B \coloneqq \{w \in W : \ell(w) = B\}.$$

Let $\succ$ denote a preference profile, and let g^{θ} be a randomized matching mechanism parameterized by θ. We write $g^{\theta}_{wf}(\succ) \in [0,1]$ for the (marginal) probability that worker w is matched to firm f under g^{θ} given $\succ$.

[1] When $n = m$ and the outside option is ignored, this reduces to a doubly stochastic matrix.

Outcome function. Fix a worker $w \in W$ and let $T_1(w, \succ) \subseteq F$ denote the set of w's most-preferred firms :

$$T_1(w, \succ) \;\coloneqq\; \arg\max_{f \in F} \; p_{wf}(\succ),$$

We define the quality-of-outcome function as the probability of attaining a best possible outcome for w:

$$\phi_1\big(w, g^\theta, \succ\big) \;\coloneqq\; \sum_{f \in F} g^\theta_{wf}(\succ)\, \mathbf{1}\{f \in T_1(w, \succ)\} \;=\; \sum_{f \in T_1(w,\succ)} g^\theta_{wf}(\succ).$$

Top-k extension. More generally, let $T_k(w, \succ) \subseteq F$ be the set of w's top-k firms under $\succ_w$. Define

$$\phi_k\big(w, g^\theta, \succ\big) \;\coloneqq\; \sum_{f \in F} g^\theta_{wf}(\succ)\, \mathbf{1}\{f \in T_k(w, \succ)\} \;=\; \sum_{f \in T_k(w,\succ)} g^\theta_{wf}(\succ).$$

Fairness violation and expected fairness. We measure fairness violation as the difference in average outcome between groups:

$$\mathrm{fai}_k\big(g^\theta, \succ\big) \;\coloneqq\; \frac{1}{|W_A|} \sum_{w \in W_A} \phi_k\big(w, g^\theta, \succ\big) \;-\; \frac{1}{|W_B|} \sum_{w \in W_B} \phi_k\big(w, g^\theta, \succ\big)\,.$$

The expected fairness (under a distribution over preference profiles $\succ$) is then

$$\mathrm{FAI}_k\big(g^\theta\big) \;\coloneqq\; \mathbb{E}_\succ\big[\mathrm{fai}_k\big(g^\theta, \succ\big)\big]\,.$$

This fairness definition can naturally be extended to more than two groups and also by defining groups in the firm side. For clarity of demonstration, we will work with two groups only in the worker side.

2.1.2 Efficiency.

In the same spirit, we define an *ordinal efficiency* criterion based on top-k allocations. For each worker $w \in W$, recall the top-k set $T_k(w, \succ) \subseteq F$ under $\succ_w$ and the associated outcome function

$$\phi_k\big(w, g^\theta, \succ\big) \;=\; \sum_{f \in T_k(w,\succ)} g^\theta_{wf}(\succ),$$

which represents the probability that w is matched to one of its top-k firms.

Analogously, for each firm $f \in F$, let $T_k(f, \succ) \subseteq W$ denote the set of f's top-k workers under $\succ_f$, and define

$$\phi_k\big(f, g^\theta, \succ\big) \;\coloneqq\; \sum_{w \in T_k(f,\succ)} g^\theta_{wf}(\succ),$$

interpreted as the probability that f is matched to one of its top-k workers.

Efficiency and expected efficiency. We aggregate both sides by averaging the corresponding outcome functions:

$$\mathrm{eff}_k\big(g^\theta, \succ\big) \;\coloneqq\; \frac{1}{2}\left(\frac{1}{|W|} \sum_{w \in W} \phi_k\big(w, g^\theta, \succ\big) \;+\; \frac{1}{|F|} \sum_{f \in F} \phi_k\big(f, g^\theta, \succ\big)\right).$$

The expected efficiency (under a distribution over preference profiles $\succ$) is then

$$\mathrm{EFF}_k\big(g^\theta\big) \;\coloneqq\; \mathbb{E}_\succ\big[\mathrm{eff}_k\big(g^\theta, \succ\big)\big]\,.$$

2.2 Benchmark matching mechanisms

We consider three benchmark mechanisms: the *deferred-acceptance* (DA) mechanism which is a stability reference point and two other mechanisms that maximize fairness and efficiency.

Benchmark 1: Worker-proposing deferred acceptance (DA). (The firm-proposing variant is defined analogously.) Each worker w maintains a set of *remaining* acceptable firms,

$$R_w \;=\; \{\, f \in F : f \succ_w \perp \,\} \setminus \{\text{firms that have already rejected } w\}.$$

Repeat the following steps until all proposals are accepted:

- **Proposals:** For every $w \in W$, if $R_w \neq \varnothing$, worker w proposes to its most-preferred firm in R_w.
- **Tentative acceptances:** For every $f \in F$, firm f tentatively accepts its most-preferred proposal (if any) and rejects all other proposals it received.
- **Updates:** For every $w \in W$, if w is rejected by firm f, then update $R_w \leftarrow R_w \setminus \{f\}$.

Benchmark 2: Efficiency-optimal matching. Fix a preference profile $\succ$ and an integer $k \geq 1$. Let $T_k(w, \succ) \subseteq F$ denote the set of worker w's top-k firms under $\succ_w$. We define the efficiency-optimal benchmark as the solution of the following assignment problem:

$$\max_{x \in \{0,1\}^{|W|\times|F|}} \quad \frac{1}{|W|} \sum_{w \in W} \sum_{f \in T_k(w,\succ)} x_{wf}$$

subject to the feasibility constraints

$$\sum_{f \in F} x_{wf} \leq 1 \quad \forall w \in W, \qquad \sum_{w \in W} x_{wf} \leq 1 \quad \forall f \in F.$$

The resulting deterministic matching can be viewed as maximizing the fraction of workers assigned to one of their top-k firms.

Benchmark 3: Fairness-optimal matching. Let $\ell : W \to \{A, B\}$ and (W_A, W_B) be as defined in the fairness section, and define the group-outcome gap for a deterministic matching $x \in \{0,1\}^{|W|\times|F|}$ as

$$\mathrm{fai}_k(x, \succ) \;\coloneqq\; \frac{1}{|W_A|} \sum_{w \in W_A} \sum_{f \in T_k(w,\succ)} x_{wf} \;-\; \frac{1}{|W_B|} \sum_{w \in W_B} \sum_{f \in T_k(w,\succ)} x_{wf}.$$

We define the fairness-optimal benchmark as any solution of

$$\min_{x \in \{0,1\}^{|W|\times|F|}} \quad \left|\mathrm{fai}_k(x, \succ)\right|$$

subject to the same feasibility constraints

$$\sum_{f \in F} x_{wf} \leq 1 \quad \forall w \in W, \qquad \sum_{w \in W} x_{wf} \leq 1 \quad \forall f \in F.$$

2.3 Two-Sided Matching as a Learning Problem

For a mechanism parameterized by g_θ, the training problem is :

$$\min_{\theta} \ \lambda \cdot |\text{fai}_k(g_\theta)| - \ (1-\lambda) \cdot \text{eff}_k(g_\theta), \tag{13}$$

where $\lambda \in [0,1]$ controls the tradeoff between fairness and efficiency. k is fixed throughout experiments.

3 Related Work

3.1 Deep Learning for Two-Sided Matching [6]

This paper builds the basis of our work and presents a deep learning method to satisfy the constraints of stability and strategy-proofness of one-to-one matchings. It first shows how stability and strategy-proofness are adapted to randomized matchings, then presents the model architecture and optimization scheme to yield the best trade-off.

Ex ante stability. A randomized matching r is said to be *ex ante stable* if it generates no *ex ante justified envy*, i.e., if neither of the following situations occurs:

1. There exist a worker $w \in W$, a firm $f \in F$, and some agents $f' \in F \cup \{\perp\}$ and $w' \in W \cup \{\perp\}$ such that

$$r_{wf'} > 0, \quad r_{w'f} > 0, \quad f \succ_w f', \quad \text{and} \quad w \succ_f w'.$$

2. There exists a worker w and a firm $f' \in F$ such that

$$r_{wf'} > 0 \quad \text{and} \quad \perp \succ_w f',$$

or there exists a firm f and a worker $w' \in W$ such that

$$r_{w'f} > 0 \quad \text{and} \quad \perp \succ_f w'.$$

Ordinal strategy-proofness. A randomized matching mechanism g is *ordinally strategy-proof* (ordinal SP) if and only if, for every agent $i \in W \cup F$, for every preference profile $\succ$, for every utility function u_i representing $\succ_i$ (i.e., any $\succ_i$-utility for agent i), and for every alternative report $\succ'_i$, we have

$$\mathbb{E}_{\mu \sim g(\succ_i, \succ_{-i})}\big[u_i(\mu(i))\big] \ \geq \ \mathbb{E}_{\mu \sim g(\succ'_i, \succ_{-i})}\big[u_i(\mu(i))\big].$$

By this definition, no worker or firm can imporve their expected utility by misreporting their preference order.

First-order stochastic dominance. A randomized matching mechanism g satisfies *first-order stochastic dominance* (FOSD) if and only if, for every worker $w \in W$, every firm $f' \in F$ such that $f' \succ_w \perp$, every report profile of the other agents $\succ_{-w}$, and every alternative report $\succ'_w$, we have

$$\sum_{f \in F:\ f \succ_w f'} g_{wf}(\succ_w, \succ_{-w}) \ \geq \ \sum_{f \in F:\ f \succ_w f'} g_{wf}(\succ'_w, \succ_{-w}). \tag{2}$$

A two-sided matching mechanism is ordinal SP if and only if it satisfies FOSD. The FOSD condition is useful when quantifying the constraint.

Surrogate functions for quantifying the constraints. For worker w and firm f, the **stability violation** at profile $\succ$ is defined as

$$stv_{wf}(g^\theta, \succ) = \left(\sum_{w' \in \overline{W}} g^\theta_{w'f}(\succ) \cdot \max\{q^\succ_{wf} - q^\succ_{w'f}, 0\} \right) \times \left(\sum_{f' \in \overline{F}} g^\theta_{wf'}(\succ) \cdot \max\{p^\succ_{wf} - p^\succ_{wf'}, 0\} \right) . \tag{1}$$

For the **strategy-proofness violation**, the FOSD is used to build the following surogate function. Let $\succ_{-i} = (\succ_1, \ldots, \succ_{i-1}, \succ_{i+1}, \ldots, \succ_{n+m})$. For a preference profile $\succ \in P$ and a mechanism g_θ, define, for any worker w and firm f,

$$\Delta_{wf}(g_\theta, \succ'_w, \succ) \;=\; g_{\theta,wf}(\succ'_w, \succ_{-w}) - g_{\theta,wf}(\succ_w, \succ_{-w}) \,.$$

The regret of worker w (respectively, firm f) is defined as

$$\text{regret}_w(g_\theta, \succ) = \max_{\succ'_w \in P} \left(\max_{f' \succ_w \perp} \sum_{f \in F:\, f \succ_w f'} \Delta_{wf}(g_\theta, \succ'_w, \succ) \right), \tag{6}$$

$$\text{regret}_f(g_\theta, \succ) = \max_{\succ'_f \in P} \left(\max_{w' \succ_f \perp} \sum_{w \in W:\, w \succ_f w'} \Delta_{wf}\big(g_\theta, \succ'_f, \succ\big) \right).$$

The average regret on profile $\succ$ is

$$\text{regret}(g_\theta, \succ) = \frac{1}{2} \left(\frac{1}{|W|} \sum_{w \in W} \text{regret}_w(g_\theta, \succ) \;+\; \frac{1}{|F|} \sum_{f \in F} \text{regret}_f(g_\theta, \succ) \right).$$

Neural Network Architecture. A randomized matching mechanism is parametrized by a feed-forward neural network

$$g_\theta : P \to \Delta(B), \qquad \theta \in \mathbb{R}^d,$$

mapping a preference profile to a distribution over matchings.

The full input is the concatenation $(p^\succ_w)_{w \in W}$ and $(q^\succ_f)_{f \in F}$, i.e., $2nm$ real numbers. Given an input profile, the network produces nonnegative scores for worker-firm pairs (including unmatched options), masks out unacceptable pairs to enforce individual rationality, and then normalizes these scores across workers and firms so that each agent's total assignment probability is at most 1. The resulting matrix of marginals is therefore weakly doubly stochastic and can be interpreted as a randomized matching; by standard decomposition results, such marginals correspond to a distribution over feasible deterministic matchings.

Training procedure For a mechanism parameterized by g_θ, the training problem is :

$$\min_\theta \;\; \lambda \cdot \text{stv}(g_\theta) \;+\; (1 - \lambda) \cdot \text{rgt}(g_\theta), \tag{13}$$

where $\lambda \in [0, 1]$ controls the tradeoff between stability and strategy-proofness.

3.2 Automated Many-to-One Matching Mechanism Design Using Deep Learning [10]

In this work, the objective is to extend the learning framework from the first paper to many-to-one matching (each school has a capacity quota of students) as well as evaluate the model on a new constraint: social welfare. Social welfare is the sum of the benefits that the agents gain in matching and is therefore defined as :

$$\text{welfare}\big(g^{\theta}_{sc}\big) = \frac{1}{2}\left(\frac{1}{n}+\frac{1}{m}\right)\sum_{s\in S}\sum_{c\in C} g^{\theta}_{sc}(\succ)\cdot(p_{sc}+q_{sc})\,.$$

where S design the set of students and C the set of schools, p and q referring to the preference order of students and schools.

The objective function then takes the following expression:

$$\min_{\theta}\ -\text{wel}(g_\theta)\ +\ \lambda\cdot\text{stv}(g_\theta)\ +\ (1-\lambda)\cdot\text{rgt}(g_\theta)$$

For many-to-one matching, the proposed framework is:

1. Construct an expanded set C' by representing each capacity quota of a school in C as a distinct agent in C'.
2. Generate students' preferences over schools, denoted $\succ_S$.
3. Build adjusted students' preferences $\succ'_S$ over C' from $\succ_S$ by expanding each school into its quota-specific agents, prioritizing higher-ranked quota positions when ordering within the same school.
4. Generate schools' preferences over students, denoted $\succ_C$.
5. Build unified preferences $\succ_{C'}$ over students for the agents in C' by lifting $\succ_C$ to the quota-specific agents of each school.

Thus, the input vector takes into account the capacity quotas and is fed to the same neural network used in [6].

4 Timeline and milestones

We organize the remaining work into five workstreams: (i) architecture design with permutation-equivariance, (ii) implementation of benchmarks and evaluation protocol, (iii) computational-complexity reductions via restricted/structured preference domains, (iv) extensions to many-to-one matching and additional constraints and (v) experiments, analysis, and final write-up. The timeline is given below.

Jan. 19 – Feb. 1. Complete the essential literature review on permutation-equivariant architectures [12], and set up the codebase so that data generation, training, and evaluation are already working end-to-end on small instances.

Feb. 2 – Feb. 15. Implement the two benchmark mechanisms (efficiency-optimal and fairness-optimal) and validate them, then develop and test at least one permutation-equivariant neural architecture for g^{θ}, producing initial fairness–efficiency trade-off results.

Feb. 16 – Mar. 1. Focus on computational improvements by restricting reports (e.g., top-k truncation) and exploring at least one structured preference domain (e.g., single-peaked [11]), while iterating on the neural mechanism to ensure stable training and meaningful improvements over benchmarks.

Mar. 2 – Mar. 15. Extend the framework to many-to-one matching with capacity constraints, replicate the main fairness–efficiency experiments in this setting, and prototype how additional desiderata such as stability or strategy-proofness could be incorporated.

Mar. 16 – Mar. 31. Run the final experimental loop to determine the final hyperparameters and overall structure, produce the definitive figures and tables, and complete the report and a cleaned reproducible code.

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